Linear Approximation and L'Hôpital's Rule

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

Compute second derivative: $f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2} \Rightarrow f''(x) = 2x^{-3} = \frac{2}{x^3}$. On $[1, 3]$, $|f''(x)| = \frac{2}{x^3}$ is decreasing, so its maximum occurs at $x = 1$: $K = \frac{2}{1^3} = 2$. Using the Trapezoidal error bound with $a = 1, b = 3$: $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{2(3-1)^3}{12n^2} = \frac{2(8)}{12n^2} = \frac{16}{12n^2} = \frac{4}{3n^2}$. Require.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

1. Identify $P(x) = \frac{2}{x}$ and $Q(x) = 4x$. 2. Calculate the integrating factor $\mu(x)$: $\mu(x) = \exp\left(\int \frac{2}{x}\,dx\right) = e^{2\ln x} = e^{\ln(x^2)} = x^2$. 3. Multiply through by x^2: $x^2\frac{dy}{dx} + 2xy = 4x^3 \Rightarrow \frac{d}{dx}[x^2 y] = 4x^3$. 4. Integrate both sides: $x^2 y = \int 4x^3\,dx = x^4 + C$. 5. Apply $y(1) = 2$: $(1)^2(2) = 1^4 + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$. 6. Explicit solution: $y = \frac{x^4+1}{x^2} = x^2 + \frac{1}{x^2}$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

1. Write the characteristic equation: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m_1 = 2,\ m_2 = 3$. 2. General solution: $y(x) = Ae^{2x} + Be^{3x}$. 3. Derivative: $y'(x) = 2Ae^{2x} + 3Be^{3x}$. 4. Apply initial conditions: $y(0) = 1 \Rightarrow A + B = 1$. $y'(0) = 3 \Rightarrow 2A + 3B = 3$. Multiply first equation by 2: $2A + 2B = 2$. Subtract from second: $B = 1$. Then $A = 1 - 1 = 0$. Particular Solution: $y(x) = e^{3x}$.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 3 · AdvancedEvaluate the definite integral analytically: $\int_{-3}^{3} \left(x^5\sqrt{9-x^2} + \sqrt{9-x^2}\right)dx$.

Split the integral using linearity: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx + \int_{-3}^{3}\sqrt{9-x^2}\,dx$. Part 1 · Symmetry Analysis: Let $g(x) = x^5\sqrt{9-x^2}$. Then $g(-x) = (-x)^5\sqrt{9-(-x)^2} = -x^5\sqrt{9-x^2} = -g(x)$. Because g is an odd function integrated over symmetric limits $[-3, 3]$: $\int_{-3}^{3} x^5\sqrt{9-x^2}\,dx = 0$. Part 2 · Geometric Area: The integrand $y = \sqrt{9-x^2} \ge 0$ represents the upper half of the circle $x^2 + y^2 = 9$ with radius $r = 3$. The integral from.

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5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

Step 1 · Volume Accumulation Function: Net rate of change is $V'(t) = R(t) - L(t) = (120 - 4t) - 3t = 120 - 7t$. Since the tank is initially empty, $V(0) = 0$: $V(t) = \int_0^t (120 - 7u)\,du = 120t - \frac{7}{2}t^2$. Step 2 · Maximum Volume: Maximum occurs when $V'(t) = 0 \Rightarrow 120 - 7t = 0 \Rightarrow t = \frac{120}{7} \approx 17.14$ hours. Since x for $t > 0$: $120t - \frac{7}{2}t^2 = 0 \Rightarrow t(120 - \frac{7}{2}t) = 0 \Rightarrow t = \frac{240}{7} \approx 34.29$ hours. Since t.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · Advanced A tank initially contains 100 liters of pure water. Brine containing 0.5 kg of salt per liter enters the tank at a rate of 4 L/min, and the well-stirred solution leaves the tank at the same rate of 4 L/min. Find the amount of salt $S(t)$ in the tank after t minutes.

Rate of change of salt: $\frac{dS}{dt} = \text{Rate In} - \text{Rate Out}$. Rate In $= (0.5 \text{ kg/L}) \times (4 \text{ L/min}) = 2 \text{ kg/min}$. Since volume remains constant at 100 L, salt concentration is $S(t)/100$. Rate Out $= \left(\frac{S(t)}{100} \text{ kg/L}\right) \times (4 \text{ L/min}) = \frac{4S}{100} = 0.04S$ kg/min. Set up the linear ODE: $\frac{dS}{dt} = 2 - 0.04S = -0.04(S - 50)$ Separate variables.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · Advanced Evaluate the improper integral or establish its divergence: $\int_0^\infty xe^{-x/2}\,dx$

Express as a limit of a definite integral: $\int_0^\infty xe^{-x/2}\,dx = \lim_{t\to\infty} \int_0^t xe^{-x/2}\,dx$ Integrate by parts on $[0, t]$: Let $u = x \Rightarrow du = dx$, and $dv = e^{-x/2}dx \Rightarrow v = -2e^{-x/2}$. $\int_0^t xe^{-x/2}\,dx = [-2xe^{-x/2}]_0^t - \int_0^t (-2e^{-x/2})\,dx = -2te^{-t/2} + 0 + 2[-2e^{-x/2}]_0^t$ $= -2te^{-t/2} - 4(e^{-t/2} - 1) = 4 - e^{-t/2}(2t + 4)$ Take the limit as $t \to \infty$: By L'Hôpital's Rule.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert the rectangular point $(0, -3)$ to polar coordinates.

Theorem (Rectangular to Polar Conversion): For rectangular coordinates (x, y), the polar radius and angle satisfy: $r = \sqrt{x^2 + y^2}$, $\tan\theta = \frac{y}{x}$ (with quadrant resolution). Step 1 · Compute radius r: $r = \sqrt{0^2 + (-3)^2} = \sqrt{0 + 9} = \sqrt{9} = 3$. Step 2 · Determine angle θ: The point $(0, -3)$ lies on the negative y-axis. Moving counterclockwise from the positive x-axis, the directional angle is $\theta = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). Conclusio.

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9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left| \frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}} \cdot \frac{n^2 4^n}{(2x-1)^n} \right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x-1=4$, giving $\sum_{n=1}^{\infty} \frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p-series test ($p=2>1$). Endpoint $x=-3/2$: $2x-1=-4$, giving.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find the speed of a particle with $x=t^2,\ y=t^2$ at $t=1$.

Theorem (Particle Kinematics & Speed Formula): For a particle whose position vector is $\vec{r}(t)=\langle x(t), y(t)\rangle$, the velocity vector is $\vec{v}(t)=\langle x'(t), y'(t)\rangle$, and its instantaneous speed is the Euclidean magnitude of velocity: $\text{Speed} = ||\vec{v}(t)|| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Step 1 · Determine velocity components: $x'(t) = \frac{d}{dt}[t^2] = 2t, \quad y'(t) = \frac{d}{dt}[t^2] = 2t$. Step 2 · Evaluate components at.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

1. Substitute $x = 2\tan\theta \Rightarrow dx = 2\sec^2\theta\,d\theta$. Then $4+x^2 = 4+4\tan^2\theta = 4\sec^2\theta$, so: $(4+x^2)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$. 2. Substitute into the integral: $\int \frac{2\sec^2\theta}{8\sec^3\theta}\,d\theta = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$. 3. Express $\sin\theta$ in terms of x: From $\tan\theta = \frac{x}{2}$, construct a right triangle with opposite side x, adjacen.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x+y+z=1$ and $\pi_2 : 2x-y+3z=4$.

1. Normal vectors: $\mathbf{n}_1 = \langle 1,1,1\rangle$, $\mathbf{n}_2 = \langle 2,\ -1, 3\rangle$. 2. Direction vector $\mathbf{d} = \mathbf{n}_1 \times \mathbf{n}_2$: $\mathbf{d} = |\mathbf{i},\mathbf{j},\mathbf{k};\ 1,1,1;\ 2,\ -1,3| = \mathbf{i}(3-(-1)) - \mathbf{j}(3-2) + \mathbf{k}(-1-2) = \langle 4;\ -1;\ -3\rangle$. 3. Find a point on the line: set $z=0$: $\{x+y=1;\ 2x-y=4 \Rightarrow 3x=5 \Rightarrow x=\frac{5}{3},\ y=1-\frac{5}{3}=-\frac{2}{3}$. P.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

1. Sort all edges in non-decreasing order of weight: BC (1), AC (2), DE (2), EF (3), AB (4), BD (5), DF (6), CD (8), CE (10). 2. Apply Kruskal's algorithm (selecting $V - 1 = 6 - 1 = 5$ edges without cycles): - Add BC (weight 1) [Tree components: {B, C}] - Add AC (weight 2) [Merges A: {A, B, C}] - Add DE (weight 2) [Component: {D, E}] - Add EF (weight 3) [Merges F: {D, E, F}] - Consider AB (weight 4): A and B are already connected in {A, B, C} (would create cycle AB-BC-CA), so SKIP AB. - Add BD (weight 5): Connects.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

First find the components: $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$. The first derivative is $\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$. Differentiating with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2} + \frac{3}{2t^2}$. Then $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$. At $t = 2$: numerator is $\frac{3}{2} + \frac{3}{8} = \frac{15}{8}$; denominator is $2(2) = 4$. Thus.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

Compute $r'(\theta) = -\sin\theta$. Form the radicand: $r^2 + (r')^2 = (1 + \cos\theta)^2 + (-\sin\theta)^2 = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta = 2(1 + \cos\theta)$. Recall the half-angle identity $1 + \cos\theta = 2\cos^2(\theta/2)$: $r^2 + (r')^2 = 2(2\cos^2(\theta/2)) = 4\cos^2(\theta/2)$. Taking the square root for $\theta \in [0, \pi]$: $\sqrt{4\cos^2(\theta/2)} = 2\cos(\theta/2)$. By symmetry across $[0, 2\pi]$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.

Compute derivatives: $\frac{dx}{dt} = 3\cos^2 t(-\sin t) = -3\cos^2 t\sin t$ $\frac{dy}{dt} = 3\sin^2 t(\cos t) = 3\sin^2 t\cos t$ Form the integrand: $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4 t\sin^2 t + 9\sin^4 t\cos^2 t = 9\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t) = 9\cos^2 t\sin^2 t$ Since $t \in [0, \pi/2]$, $\cos t \ge 0$ and $\sin t \ge 0$, so the square root is: $\sqrt{9\cos^2 t\sin^2 t} = 3\cos t\sin t = \frac{3}{2}\sin(2t)$ Integrate on $[0, \pi/2]$.

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 6.01 Antiderivatives and Riemann Sums**

Using the binomial series for $(1-x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x)=\arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1+u)^p = 1+pu+\frac{p(p-1)}{2!}u^2+\ldots$. Here $u=-x^2$ and $p=-1/2$: $(1-x^2)^{-1/2} = 1+(-\frac{1}{2})(-x^2)+\frac{(-1/2)(-3/2)}{2}(-x^2)^2+O(x^6) = 1+\frac{1}{2}x^2+\frac{3}{8}x^4+O(x^6)$. 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, integrate term-by-term: $\arcsin x = \int_0^x (1+\frac{1}{2}t^2+\frac{3}{8}t^4+\ldots)dt = x+\frac{1}{6}x^3+\frac{3}{40}x^5+\ldots$ 3. The binomial series converge.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x)=x+\frac{1}{x}$ on $[1,4]$, and find all values of c that satisfy the theorem.

Step 1 · Check Hypotheses: $f(x)$ is continuous on $[1,4]$ and differentiable on $(1,4)$ since the only discontinuity $x=0$ lies outside $[1,4]$. The hypotheses of MVT hold. Step 2 · Compute Average Rate of Change: $\frac{f(4)-f(1)}{4-1} = \frac{(4+1/4)-(1+1)}{3} = \frac{17/4-2}{3} = \frac{9/4}{3} = \frac{3}{4}$. Step 3 · Solve $f'(c)=\frac{3}{4}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Level 2 · MediumEvaluate $\int x^3\sin(2x)\,dx$ using the Tabular Method.

Tabulate signs, derivatives of x^3, and antiderivatives of $\sin(2x)$: • $(+)$ x^3 $-\frac{1}{2}\cos(2x)$ • $(-)$ $3x^2$ $-\frac{1}{4}\sin(2x)$ • $(+)$ $6x$ $\frac{1}{8}\cos(2x)$ • $(-)$ 6 $\frac{1}{16}\sin(2x)$ • $(+)$ 0 $-\frac{1}{32}\cos(2x)$ Sum the diagonal products: $= -\frac{1}{2}x^3\cos(2x)+\frac{3}{4}x^2\sin(2x)+\frac{3}{4}x\cos(2x)-\frac{3}{8}\sin(2x)+C$. Group terms.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 8.01 Area Between Curves**

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

Method · Taylor/Maclaurin Polynomial Approximation: Step 1 · Define the two-term (degree 1) Maclaurin polynomial for e^x: $P_1(x)=1+x$. Step 2 · Evaluate at $x=0.1$: $P_1(0.1)=1+0.1=\mathbf{1.1}$. Step 3 · Error analysis via Lagrange Remainder: By Taylor's Theorem: $|R_1(0.1)| = \left|\frac{f''(c)}{2!}(0.1)^2\right| = \frac{e^c}{2}(0.01)$ for some $c\in(0,0.1)$. Since $e^c < e^{0.1} < 1.2$, $|R_1(0.1)| < \frac{1.2}{2}(0.01) = 0.006$. Exact value: $e^{0.1}\approx 1.10517$. The tw.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the local linear approximation $L(x)$ of $f(x) = \sqrt{x}$ at $x = 25$?

a. $L(x) = 5 + \frac{1}{10}(x - 25)$ ✓

b. $L(x) = 5 + \frac{1}{5}(x - 25)$

c. $L(x) = 5 + \frac{1}{2}(x - 25)$

d. $L(x) = \frac{1}{10}x$

$f(25) = 5$. $f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(25) = \frac{1}{10}$. Linearization: $L(x) = f(a) + f'(a)(x - a) = 5 + \frac{1}{10}(x - 25)$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Using the linearization of $f(x) = \sqrt{x}$ at $a = 25$, approximate $\sqrt{26}$.

a. 5.10 ✓

b. 5.05

c. 5.20

d. 5.01

$L(26) = 5 + 0.1(26 - 25) = 5 + 0.1 = 5.10$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Is the tangent line approximation for $f(x) = \sqrt{x}$ an overestimate or underestimate, and why?

a. **An overestimate because $f''(x) < 0$ (concave down)** ✓

b. An underestimate because $f''(x) > 0$ (concave up)

c. Exact because f is monotonic

d. Depends on the step size

Since $f''(x) = -\frac{1}{4}x^{-3/2} < 0$, the curve is concave down and lies below its tangent lines, making tangent line approximations overestimates.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which requirement MUST be verified before applying L'Hôpital's Rule to $\lim_{x \to c} \frac{f(x)}{g(x)}$?

a. The limit produces an indeterminate form $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ ✓

b. $f(c) = 0$ only

c. $f'(c) = 0$

d. $g(x) > 0$ for all x

L'Hôpital's Rule applies ONLY when the quotient evaluates to an indeterminate form of type $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

As $x \to \infty$, which of the following functions grows the fastest?

a. e^x ✓

b. x^{10}

c. $x \ln x$

d. 2^x

Exponentials with larger base grow faster than any polynomial or power function. Since $e \approx 2.718 > 2$, e^x grows fastest.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$, what does this conclude about their growth rates?

a. $f(x)$ **grows faster than** $g(x)$ ✓

b. $g(x)$ grows faster than $f(x)$

c. They grow at the same rate

d. Neither function grows

By definition of relative growth rates, $\lim_{x \to \infty} \frac{f(x)}{g(x)} = \infty$ indicates that $f(x)$ grows strictly faster than $g(x)$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The midpoint rule approximation for $\int_0^1 \int_0^1 f(x, y)\, dA$ with one rectangle:

a. $f(1, 1)$

b. $f(0, 0)$

c. $f(1/2, 1/2)$ ✓

d. $f(0, 1)$

Midpoint rule: sample at midpoint $(1/2, 1/2)$.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For $\frac{dy}{dx} = y,\ y(0) = 1$ (exact solution $y = e^x$), a single Euler step with $h = 0.25$ gives y_1. Compared with the exact value $y(0.25) = e^{0.25}$, Euler's approximation:

a. overestimates the true value

b. equals the true value exactly

c. cannot be compared

d. underestimates the true value ✓

Euler gives $y_1 = 1.25$; true value $e^{0.25} \approx 1.2840$. Since $y = e^x$ is concave up, the tangent-line step underestimates the curve.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Using L'Hôpital's rule: $\lim_{x \to 0} \frac{\sin 3x}{5x} =$

a. 3/5 ✓

b. 5/3

c. 0

d. 1

Applying Limits and Continuity principles, $\cos 3x \cdot 3/(5) \to 3/5$. Therefore, the verified correct option is 3/5.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

L'Hôpital advanced: $\lim_{x \to 0^+} x^x =$

a. 1 ✓

b. 0

c. e

d. DNE

$x \ln x \to 0 \Rightarrow e^{x \ln x} \to e^0 = 1$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use L'Hôpital's rule: $\lim_{x \to 0} \frac{e^x - 1}{\sin x}$ equals:

a. 0

b. **1** ✓

c. e

d. undefined

Differentiating num/denom: $e^x / \cos x|_{x=0} = 1/1 = 1$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

L'Hôpital's rule applies to $\lim_{x \to a} \frac{f}{g}$ when the limit is of the form:

a. 0/1

b. 1/0

c. 0/0 ✓

d. 1/1

Indeterminate form $0/0$ or ∞/∞.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Best linear approximation to $e^x \sin x$ at $x = 0$:

a. x ✓

b. $1 + x$

c. $2x$

d. $x + x^2$

$f(0) = 0$, $f'(0) = e^0(\sin 0 + \cos 0) = 1$; $L = x$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

LIATE rule for IBP: first choice for u among x^2, e^x, $\ln x$:

a. e^x

b. x^2

c. $\ln x$ ✓

d. any of them

Logarithm comes before Algebraic: $u = \ln x$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Linear approximation: $f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$. This is accurate near:

a. (a, b) **if** f **is differentiable** ✓

b. far from (a, b)

c. only when f is quadratic

d. when $ablaf = 0$

Linearisation is accurate near the point.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the quadratic function $f(x) = x^2 - 4x + 3$ and the linear function $g(x) = x - 1$. The points of intersection are given by their solutions. What is the y-coordinate of the point of intersection where x = 1?

a. **0** ✓

b. 1

c. -1

d. 2

To find the points of intersection, we set $f(x) = g(x)$: $x^2 - 4x + 3 = x - 1$. $x^2 - 5x + 4 = 0$. $(x - 1)(x - 4) = 0$. The x-coordinates of the intersection points are $x = 1$ and $x = 4$. We are asked for the y-coordinate when $x = 1$. We can use either function to find this y-coordinate. Using $g(x) = x - 1$, when $x = 1$, $g(1) = 1 - 1 = 0$. Therefore, the y-coordinate is 0.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$ ✓

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

The integrating factor is $\mu(x) = e^{-\int \frac{1}{x} dx} = \frac{1}{x}$. Then $\left(\frac{y}{x}\right)' = x \Rightarrow \frac{y}{x} = \frac{x^2}{2} + C \Rightarrow y = \frac{1}{2}x^3 + Cx$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Probability: $f(x, y) = 6xy$ on $0 \le x \le 1, 0 \le y \le \sqrt{1 - x^2}$ (quarter circle). $P(X + Y \le 1) =$

a. $\frac{3}{4}$

b. $\frac{1}{2}$

c. $\frac{1}{4}$ ✓

d. $\frac{2}{3}$

Integrating over the triangle $x \ge 0, y \ge 0, x + y \le 1$ gives $\int_0^1 \int_0^{1-x} 6xy \, dy \, dx = 3\int_0^1 x(1 - x)^2 \, dx = 3\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{1}{4}$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2}+\frac{e^4}{2}$ ✓

b. $-1+e^4$

c. $\frac{1}{2}+\frac{e^4}{2}$

d. $-\frac{1}{4}+\frac{e^4}{4}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^2 \int_0^x e^{x^2}\,dy\,dx$, which evaluates to $-\frac{1}{2}+\frac{e^4}{2}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Area Between Curves

Use $x=u+v, y=u-v$ to compute $\iint_D (x-y)e^{x+y}\,dA$ where D is the square $|x-y|\le 1, |x+y|\le 1$:

a. 0 ✓

b. $2(e-e^{-1})$

c. $(e-1)^2$

d. $4(e-e^{-1})/e$

Under the change of variables $2u=x+y$ and $2v=x-y$, the integrand $2v\,e^{2u}$ is odd in v over the symmetric interval $[-1/2, 1/2]$, so the double integral equals 0.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$ ✓

d. $-\frac{3}{2}$

$x^2+y^2=10^2$. At $x=6$: $y=\sqrt{10^2-6^2}=8$. Differentiating: $2x\frac{dx}{dt}+2y\frac{dy}{dt}=0 \Rightarrow \frac{dy}{dt}=-\frac{x}{y}\frac{dx}{dt}=-\frac{3}{4}$ m/s.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Limits Graphically and Numerically

Iterated integral over tetrahedron $x+y+z\le 1, x,y,z\ge 0$: volume = $1/6$. The integral $\iiint x\,dV =$

a. $1/6$

b. $1/24$ ✓

c. $1/12$

d. $1/8$

By symmetry $\iiint x = \iiint y = \iiint z$; and $\iiint (x+y+z) = 3\iiint x$; by integration $\iiint (x+y+z)dV = 1/8$... actually $\iiint x\,dV = 1/24$ (standard result).

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$ ✓

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

Applying Techniques of Integration principles, $x = (x-1) + 1 \Rightarrow A = 1, B = 1$. Therefore, the verified correct option is $\frac{1}{x-1} + \frac{1}{(x-1)^2}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 The Definition of the Derivative

Evaluate $\iiint_B (x^2 + y^2 + z^2)\,dV$ where B is the ball $x^2 + y^2 + z^2 \le 1$.

a. $\frac{4\pi}{5}$ ✓

b. $\frac{4\pi}{3}$

c. $\frac{2\pi}{5}$

d. $\frac{8\pi}{5}$

In spherical coordinates the integrand is ρ^2, so $\int_0^{2\pi}\int_0^{\pi}\int_0^{1} \rho^4 \sin\phi \, d\rho\, d\phi\, d\theta = \frac{4\pi}{5}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

If $f'(x) = (x-3)$, on which interval is f increasing?

a. $(3, \infty)$ ✓

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

Applying Application of Derivatives principles, $f'(x) > 0$ when $x - 3 > 0$, i.e. $x > 3$. Therefore, the verified correct option is $(3, \infty)$.

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